A Method and System for High-Radix Physical Layer Data Encoding and Representation

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Title: A Method and System for High-Radix Physical Layer Data Encoding and Representation

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Abstract

A method and system for encoding, transmitting, and decoding data at the physical layer using a high-radix, post-binary scheme. The method involves defining a set of N discrete, distinguishable physical states on a transmission or storage medium, where N is an integer greater than two. A direct, one-to-one mapping is established between a set of abstract symbols and the N physical states. A transmitter generates a sequence of these physical states to represent a corresponding sequence of symbols. A receiver detects the sequence of states and, using the inverse mapping, decodes it back into the abstract symbols. This approach replaces binary encoding with a native, high-radix symbolic representation, independent of the specific physical medium used.

Background of the Invention

The foundation of modern digital computing and networking is the binary system. Information is encoded into a series of bits, which are physically represented by two distinct states (e.g., high/low voltage, on/off light). This binary approach provides a robust defense against noise and signal degradation, as only two states need to be reliably distinguished. However, representing complex information requires long sequences of these bits, and all higher-level data structures must ultimately be serialized into this two-state system, creating an inherent bottleneck and a layer of abstraction between the information's meaning and its physical representation.

Summary of the Invention

The present invention provides a method and system for a high-radix physical data layer that transcends the limitations of binary encoding. The invention is not a specific implementation, but a generalized framework for post-binary data representation.

The core of the method is the establishment of a "distinguishable state space" on a physical medium. This is a set of N discrete physical states (where N is greater than 2) that a system can reliably generate and differentiate. For example, these states could be N distinct wavelengths of light, N discrete voltage levels, or N unique polarization angles.

An encoding map is defined that creates a direct, one-to-one correspondence between a set of abstract symbols and the N physical states. These symbols can represent numbers (e.g., the symbol for '25' maps to a specific wavelength), mathematical operators ('add' maps to another), or any other data primitive.

A transmitter, configured with this map, encodes a sequence of symbols by generating the corresponding sequence of physical states on the medium. A receiver, also configured with the map, detects this sequence of physical states, distinguishes one from another, and decodes them back into the original abstract symbols. This allows for higher data density and a more direct representation of information than is possible with a binary system, and is defined in a manner that is agnostic to the underlying physical medium, ensuring its applicability to future technological advancements.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental, defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram of the high-radix encoding and decoding system, illustrating the mapping between abstract symbols and physical states.
FIG. 2 is a flowchart illustrating the technology-agnostic method for establishing and using a high-radix physical layer protocol.

Detailed Description of the Invention

Referring to **FIG. 1**, the system (100) for high-radix physical layer encoding is shown. A set of abstract symbols (110) is defined, including, for example, numerical values and operators. A Symbol-to-State Encoding Map (120) establishes a direct, one-to-one correspondence between each symbol and a unique physical state from a defined Distinguishable State Space (130). A transmitter (140), upon receiving a sequence of symbols, consults the map and generates a corresponding sequence of physical states (150) on a generic physical medium. A receiver (160) detects this sequence, and using the inverse of the encoding map, decodes the states back into the original sequence of abstract symbols.

FIG. 2 is a flowchart (200) of the generalized method. The first step is to define a set of N discrete physical states on a medium that are reliably distinguishable by a sensor or detector, where N ¿ 2 (210). Next, a one-to-one map is established between a set of abstract symbols and the N physical states (220). To transmit data, a sequence of abstract symbols is taken as input (230). For each symbol in the sequence, the corresponding physical state is looked up in the map and generated on the medium by a transmitter (240). A receiver then detects the sequence of physical states (250). For each detected state, the corresponding abstract symbol is looked up and the original sequence is reconstructed (260). This method is independent of how the N states are physically realized.

Claims

What is claimed is:

1. A method for representing data, comprising:

   - defining a set of N discrete physical states on a medium, wherein N is an integer greater than two, and wherein each state is reliably distinguishable from every other state in the set by a detector;
   - establishing a one-to-one mapping between a set of abstract symbols and the set of N discrete physical states;
   - encoding a sequence of said abstract symbols into a corresponding sequence of said physical states based on the established mapping; and
   - causing a transmitter to generate said sequence of physical states on the medium.

2. The method of claim 1, further comprising:

   - detecting, with a receiver, the sequence of physical states on the medium; and
   - decoding the detected sequence of physical states back into the sequence of abstract symbols using the inverse of the established mapping.

3. The method of claim 1, wherein the set of abstract symbols includes numerical values.

4. The method of claim 1, wherein the set of abstract symbols includes mathematical operators.

5. The method of claim 1, wherein the medium is a guided medium or free space.

6. The method of claim 5, wherein the discrete physical states are distinct wavelengths or polarization states of electromagnetic radiation.

7. The method of claim 1, wherein the medium is an electrical conductor and the discrete physical states are distinct voltage or current levels.

8. A system for data representation, comprising:

   - a memory storing a one-to-one mapping between a set of N abstract symbols and a corresponding set of N discrete, distinguishable physical states, where N is an integer greater than two;

- a transmitter configured to access the mapping to encode a sequence of abstract symbols into a sequence of physical states and to generate said sequence of physical states on a medium; and
- a receiver configured to detect a sequence of said physical states on the medium and to access the mapping to decode it back into the sequence of abstract symbols.

9. The system of claim 8, wherein the receiver is configured to distinguish between any two of the N physical states with a probability of error below a predetermined threshold.

Drawings

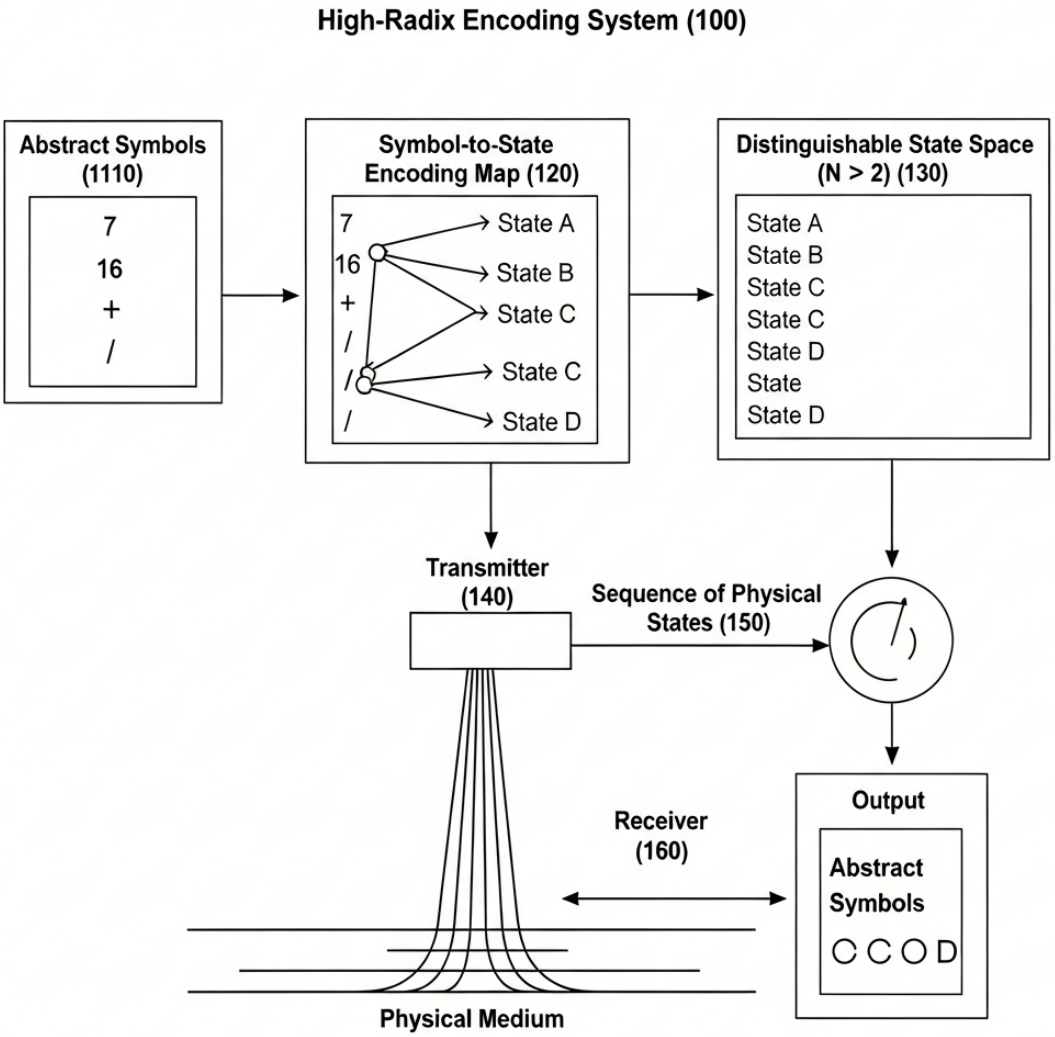


Figure 1: Conceptual diagram of the high-radix encoding and decoding system, illustrating the mapping between abstract symbols and physical states.

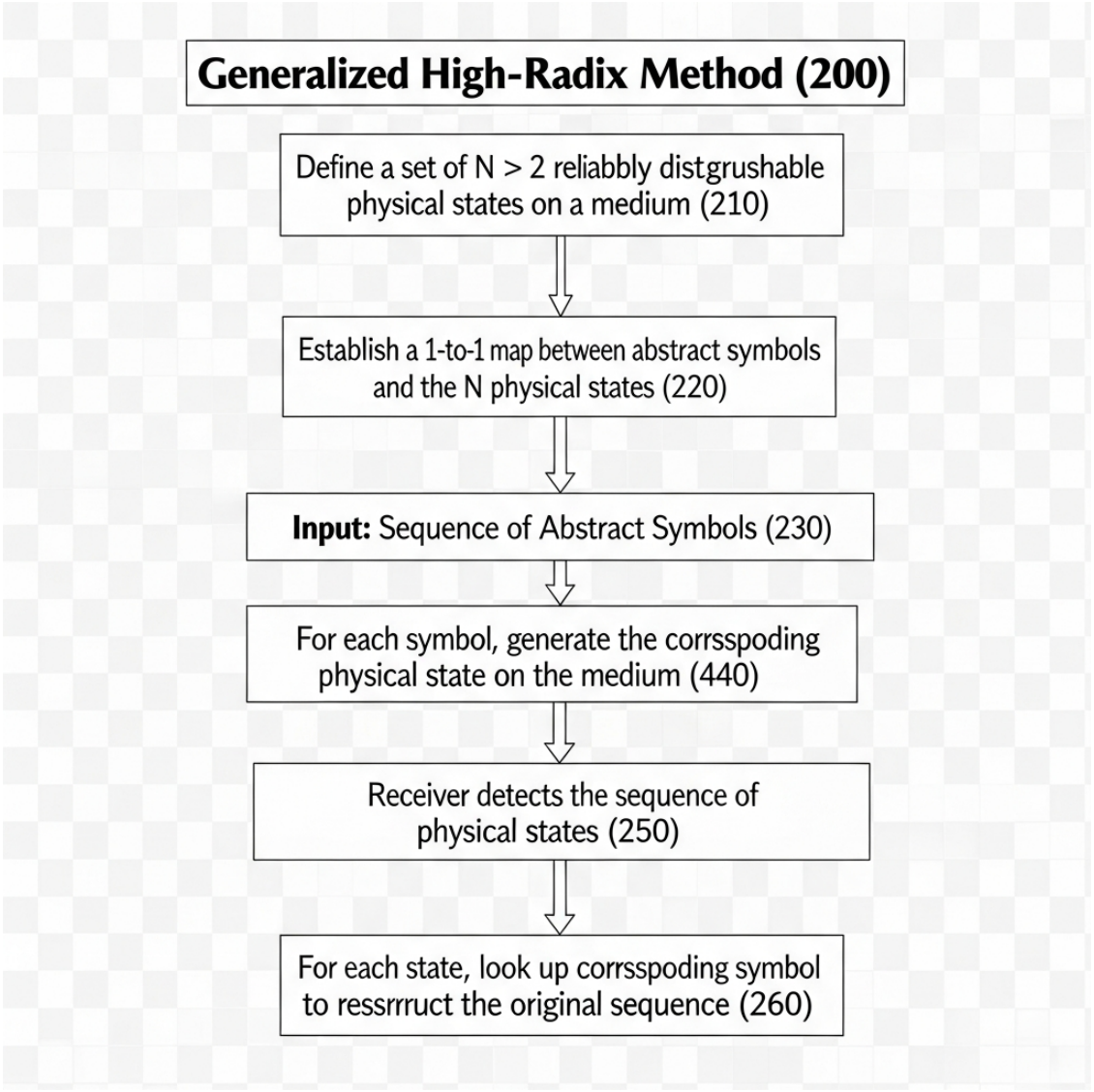


Figure 2: Flowchart illustrating the technology-agnostic method for establishing and using a high-radix physical layer protocol.